To: The Guest Editors, Omega 55th Anniversary Special Issue "Data-Driven Optimization for Smart Cities"

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Dear Professor Pan and Guest Editors,

Please consider the enclosed manuscript, "City charges for competing delivery carriers: exact best responses under probability-box travel times and leader pricing without equilibrium search", for the special issue.

A city that charges or subsidises deliveries has to predict how competing carriers react. Each reaction is an NP-hard routing problem under uncertain travel times, and the carriers interact through a congestion game. The manuscript gives one algorithm for each layer. Budget-resource labelling prices routes exactly when travel times are probability boxes with unknown dependence, and with branch-and-price it solves each carrier's robust routing problem to proven optimality on nine metropolitan instances with 22 customer zones, and with 30 to 40 customers in London and New York. Dual implementation pricing computes zone charges that make a social target an equilibrium using one best response per carrier. Aggregate-response pricing sets charges from observed zone totals alone, with a contraction rate that does not depend on instance size in the convexified game. On nine cities built from OpenStreetMap and gridded population, dual implementation pricing reached 1.07 to 1.23 times the gain of the best grid point with 5.7 to 8.3% of its routing calls in the three cities with ten seeds. The paper also gives existence and complexity results for the carriers' game and shows that policy directions persist across carrier criteria while levels do not.

The work fits the call through urban logistics and vehicle routing, robust optimisation under data uncertainty, and public--private coordination. It is a methods paper with computation on open data; it uses no machine learning, digital twin or real-time control and claims nothing in those directions. Section 1 states the novelty level of each component and Section 9 the limits of the stylised instances.

The manuscript is original, unpublished and not under review elsewhere. Code, instance pipeline, data and table-generating scripts are at https://github.com/vinhqdang/Competing-Carriers-under-Uncertainty.

Yours sincerely,

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